MA2031 EndSem Solutions

A1. Using Gauss-Jordan elimination, determine the values of $\alpha \in \mathbb{R}$ so that the linear system $\quad x + y + \alpha z = 1, \quad x - y - z = 2, \quad 2x + y - 2z = 3 \quad$ has
(a) no solutions (b) a unique solution (c) infinitely many solutions. [5]

Sol: We take the first equation as the third equation. Using Gauss-Jordan elimination, we get

$$\left[\begin{array}{ccc|c} 1 & -1 & -1 & 2 \\ 2 & 1 & -2 & 3 \\ 1 & 1 & \alpha & 1 \end{array}\right] \xrightarrow{\text{RREF}} \left[\begin{array}{ccc|c} \boxed{1} & 0 & -1 & 5/3 \\ 0 & \boxed{1} & 0 & -1/3 \\ 0 & 0 & \alpha+1 & -1/3 \end{array}\right]$$

(a) So, the system has no solutions when $\alpha = -1$.
(b) The system has a unique solution when $\alpha \neq -1$.
(c) The system does not have infinitely many solutions for any value of α.

A2. Determine $[A]_{E,E}$, the coordinate matrix of A with respect to the ordered basis E, where $A = \begin{bmatrix} 2 & 1 & 3 \\ 3 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$ and $E = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$. [5]

Sol: Form the matrix P by taking the basis vectors as its columns. Then, $[A]_{E,E} = P^{-1}AP$. Now,

$$P = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

Then, $P^{-1} = \frac{1}{2} \begin{bmatrix} 1 & 1 & -1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{bmatrix}$.

$$[A]_{E,E} = P^{-1}AP = \frac{1}{2} \begin{bmatrix} 7 & 9 & 6 \\ -1 & 1 & 2 \\ 1 & 1 & 0 \end{bmatrix}.$$

<u>Aliter</u>: Form the matrix P by taking the basis vectors as its columns.
Then, $[P|AP] \xrightarrow{\text{RREF}} [I|[A]_{E,E}]$. Now,

$$AP = \begin{bmatrix} 3 & 5 & 4 \\ 4 & 5 & 3 \\ 0 & 1 & 1 \end{bmatrix}$$

$$[P|AP] \xrightarrow{\text{RREF}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 7/2 & 9/2 & 3 \\ 0 & 1 & 0 & -1/2 & 1/2 & 1 \\ 0 & 0 & 1 & 1/2 & 1/2 & 0 \end{array}\right].$$

Hence, $[A]_{E,E} = \frac{1}{2} \begin{bmatrix} 7 & 9 & 6 \\ -1 & 1 & 2 \\ 1 & 1 & 0 \end{bmatrix}$.

A3. The matrix $A = \begin{bmatrix} 1 & -1 & 4 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{bmatrix}$ has two eigenvalues 1 and 3 with corresponding eigenvectors as $\begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$.

(a) Determine the other eigenvalue and its corresponding eigenvector. [3]

(b) Using the eigenvalues and the eigenvectors, determine $A^5 \begin{bmatrix} -2 \\ 5 \\ 2 \end{bmatrix}$. [3]

Sol: (a) The other eigenvalue is $\text{trace}(A) - 1 - 3 = 2 - 4 = -2$.
For an eigenvector, we solve the equations $(A + 2I)v = 0$, or,
$3v_1 - v_2 + 4v_3 = 0, 3v_1 + 4v_2 - v_3 = 0, 2v_1 + v_2 + v_3 = 0$. It gives $v_1 = -v_3$, $v_2 = v_3$.
So, an eigenvector is $[-1, 1, 1]^t$.
(b) Now, $[-2, 5, 2]^t = [-1, 4, 1]^t + [-1, 1, 1]^t$.
So, $A^5[-2, 5, 2]^t = 1^5[-1, 4, 1]^t + (-2)^5[-1, 1, 1]^t = [31, -28, -31]^t$.

A4. Using Gram-Schmidt orthogonalization, compute rank(A), where [4]

$$A = \begin{bmatrix} 2 & -1 & -1 & 0 \\ 1 & 0 & 1 & -4 \\ 0 & 1 & -1 & -4 \end{bmatrix}.$$

Sol: Taking columns as vectors, we get

$$v_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix}, \quad v_3 = \begin{bmatrix} -2 \\ 4 \\ -2 \end{bmatrix}, \quad v_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}. \quad \text{So, rank}(A) = 3.$$

<u>Aliter</u>: Consider the row vectors to get
$v_1 = [2, -1, -1, 0]$, $\quad v_2 = [4, 1, 7, -24]$, $\quad v_3 = [0, 1, -1, -4] - \frac{90}{641}[4, 1, 7, -24] \neq 0$.
So, rank$(A) = 3$.

A5. Determine the best approximation of $(1, 2, 1)$ from $U = \{(a, b, c) \in \mathbb{R}^3 : a+b+c = 0\}$. [3]

Sol: The best approximation (a, b, c) of $(1, 2, 1)$ from U satisfies
(i) $(a, b, c) \in U$ and (ii) $[(1, 2, 1) - (a, b, c)] \perp U$.
A basis for U is $\{(1, -1, 0),\ (1, 0, -1)\}$.
Thus, the best approximation satisfies $(1 - a, 2 - b, 1 - c) \cdot (1, -1, 0) = 0$,
$(1 - a, 2 - b, 1 - c) \cdot (1, 0, -1) = 0$ and $a + b + c = 0$.
Solving these we find that $a = -1/3$, $b = 2/3$, $c = -1/3$.
Hence, the best approximation is $(-1/3, 2/3, -1/3)$.

A6. Find the singular value decomposition of $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$. [8]

Sol: Singular values are $s_1 = 5$, $s_2 = 3$. (one may also write $s_3 = 0$.)

For $s_1^2 = 25$, $v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$; For $s_2^2 = 9$, $v_2 = \frac{1}{\sqrt{18}} \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}$;

For $s_3^2 = 0$, $v_3 = \frac{1}{3} \begin{bmatrix} 2 \\ -2 \\ -1 \end{bmatrix}$.

$u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $u_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

$$A = P\Sigma Q^* = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{18} & -1/\sqrt{18} & 4/\sqrt{18} \\ -2/3 & -2/3 & -1/3 \end{bmatrix}.$$

B7. Let $\{v_1, \ldots, v_n\}$ be a given basis of $\mathbb{R}^{n\times 1}$. Let $A \in \mathbb{R}^{n\times n}$. Prove that A is invertible if and only if $Ax = v_i$ is consistent for each $i \in \{1, \ldots, n\}$. [3]

Sol: Let A be invertible. Then, a solution of $Ax = v_i$ is $x = A^{-1}v_i$. So, $Ax = v_i$ is consistent for each i.

Conversely, suppose for each i, $Ax = v_i$ is consistent. So, let u_i be a solution of $Ax = v_i$. Then, $A[u_1, \cdots, u_n] = [v_1, \cdots, v_n]$. Since $\{v_1, \ldots, v_n\}$ is a basis of $\mathbb{R}^{n\times 1}$, the matrix $[v_1, \cdots, v_n]$ is invertible. Thus, $A[u_1, \cdots, u_n][v_1, \cdots, v_n]^{-1} = I$. As A is a square matrix, its inverse is $[u_1, \cdots, u_n][v_1, \cdots, v_n]^{-1}$.

B8. Let $A \in \mathbb{R}^{3\times 3}$ have all its entries equal to 1. Determine a matrix P so that $P^{-1}AP$ is a diagonal matrix. [3]

Sol: $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$. It has eigenvalues $3, 0, 0$.

Corresponding linearly independent eigenvectors are
$[1, 1, 1]^t$, $[1, -1, 0]^t$ and $[1, 0, -1]^t$.

Hence the matrix $P = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$ is a diagonalizing matrix.

B9. Let A be a nonzero matrix of rank 1. What are its eigenvalues? [3]

Sol: Let $A \in \mathbb{C}^{n\times n}$. $\text{rank}(A) = 1$ implies $\text{null}(A) = n - 1$.
That is, $Ax = 0$ has $n - 1$ linearly independent solutions;
these are eigenvectors for the eigenvalue 0.
Then, the other eigenvalue is $\text{trace}(A)$.

B10. Let $A \in \mathbb{R}^{5\times 5}$. Suppose $\text{Sol}(A, 0) = \{[a, b, c, d, e]^t \in \mathbb{R}^{5\times 1} : a + d + e = 0 = b + c\}$ and $\text{rank}(A - I) = 3$. Determine whether A is diagonalizable. [3]

Sol: Geometric multiplicity of eigenvalue 1 is $\text{null}(A - 1 \cdot I) = 5 - 3 = 2$.
Geometric multiplicity of eigenvalue 0 is $\text{null}(A - 0 \cdot I) = \dim(\text{Sol}(A, 0)) = 3$.
Their sum is 5 which is the order of the matrix. Hence, A is diagonalizable.

B11. Let $A \in \mathbb{C}^{n\times n}$ be a normal matrix. Let $\lambda \in \mathbb{C}$ be an eigenvalue of A associated with an eigenvector $v \in \mathbb{C}^{n\times 1}$. Show that $\overline{\lambda}$ is an eigenvalue of A^* associated with the eigenvector v. [3]

Sol: Let the eigenvalue of A corresponding to the eigenvector v by λ.
Then, $\|Av - \lambda v\|^2 = 0$.
Its expansion gives $0 = v^*A^*Av - \overline{\lambda}v^*Av - \lambda v^*Av + |\lambda|^2 v^*v$.
Use ${}^*A = AA^*$ and rearrange the RHS to get $\|A^*v - \overline{\lambda}v\|^2$.
Hence, $\|A^*v - \overline{\lambda}v\|^2 = 0$ so that v is an eigenvector of A^*.

Aliter: Since A is normal, there exists unitary P such that $A = PDP^*$.
Assume that λ occurs as the ith diagonal entry in D and the corresponding eigenvector v is the ith column of P. Now, $A^* = PD^*P^*$ says that $\overline{\lambda}$ is the ith diagonal entry of D^* and the corresponding eigenvector is the ith column of P, that is, v. Hence, v is an eigenvector associated with the eigenvalue $\overline{\lambda}$ of A^*.

B12. Let a, b, c, d be distinct real numbers. Let u, v, w be linearly independent vectors in $\mathbb{R}^{5\times 1}$. Determine whether the vectors $u + av$, $v + bw$, $w + cu$ and $u - v - dw$ are linearly independent or linearly dependent. [2]

Sol: Let $U = \text{span}\{u, v, w\}$. Since u, v, w are linearly independent, $\dim(U) = 3$.
If a vector in the list $u + av$, $v + bw$, $w + cu$, $u - v - dw$ is repeated, then they are linearly dependent. Otherwise, they are four vectors in U, which is a 3-dimensional subspace of $\mathbb{C}^{5\times 1}$; so they are linearly dependent.
In any case, the given vectors are linearly dependent.

B13. Construct a matrix $A \in \mathbb{R}^{4\times 4}$ where no column is a linear combination of any two other columns, but $\det(A) = 0$. [2]

Sol: Take $A = [e_1,\ e_2,\ e_3,\ e_1 + e_2 + e_3]$. Now, no column is a linear combination of other two columns. But $\det(A) = 0$.

B14. Let $A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$. Does there exist an invertible matrix P such that $P^{-1}AP$ is a diagonal matrix? Justify your answer. [2]

Sol: Suppose such a matrix P exists. The eigenvalues of A are $1, 1, 1, 1$. So, $P^{-1}AP = I$. Now, $P^{-1}(A - I)P = P^{-1}AP - I = 0$ has rank 0. But $A - I$ has rank 1. This contradicts the fact that the ranks of two similar matrices are equal.
Hence, there does not exist such a matrix P.

B15. Let A be a square matrix with characteristic polynomial $(t-1)^2(t-2)^3(t-3)^4(t-4)$. Determine the rank of A. [2]

Sol: $\det(A) =$ product of its eigenvalues $= 1^2 \cdot 2^3 \cdot 3^4 \cdot 4 \neq 0$. So, A is invertible.
Hence, A has rank $=$ its order $= 2 + 3 + 4 + 1 = 10$.

B16. Let $V = \left\{ (a, b, c) \in \mathbb{R}^3 : \begin{bmatrix} 7 & a & b \\ 0 & 2 & c \\ 0 & 0 & 3 \end{bmatrix} \text{ is diagonalizable} \right\}$.

Determine whether V is a subspace of $\mathbb{R}^3$. [2]

Sol: The given matrix there is upper triangular; its diagonal entries are its eigenvalues. They are distinct. So, the matrix is diagonalizable for all values of $a, b.c$. Hence, $V = \mathbb{R}^3$. It is a subspace of $\mathbb{R}^3$.

B17. Let $A \in \mathbb{C}^{10\times 10}$ have the characteristic polynomial $(t-1)^7(t-2)^2(t-3)$.
If $\text{rank}(A - I) = 5$, then determine whether A is diagonalizable or not. [2]

Sol: The geometric multiplicity of the eigenvalue 1 is $\text{null}(A - I) = 10 - 5 = 5$. The algebraic multiplicity of 1 is $7 > 5$. So, A is not diagonalizable.

B18. Let $A \in \mathbb{R}^{3\times 3}$, $A \neq 0$ but $A^2 = 0$. Determine whether A is diagonalizable. [2]

Sol: If $\lambda \in \sigma(A)$, then $Av = \lambda v$ implies $0 = A^2 v = \lambda^2 v$ implies $\lambda = 0$.
That is, the only eigenvalue is 0.
If A is diagonalizable, then $A = P^{-1} 0 P = 0$, a contradiction.
Hence, A is not diagonalizable.
